QuizTime Test Plan

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Overview and scope

This plan covers end-to-end testing of QuizTime, a single-page, five-question multiple-choice quiz, using Playwright against the app served locally by Live Server.

The app has three screens, switched by an `active` CSS class:

- Start screen: title, tagline and a Start Quiz button
- Quiz screen: question, counter ("Question X of 5"), score, four answer buttons and a progress bar
- Result screen: final score out of 5, a message based on the percentage, and a Restart Quiz button

In scope: screen transitions, question and answer rendering, scoring, correct and incorrect feedback, answer locking, auto-advance, progress bar, result messages, restart, and basic responsive layout.

Out of scope: cross-browser visual pixel comparison, performance and load testing, accessibility audits beyond basic role checks, and any backend (the app has none).

Test approach and tooling

All tests are automated UI tests written in Playwright (`@playwright/test`), each starting from a fresh page load so no test depends on another.

- **Locators:** use IDs for fixed elements (`#score` , `#current-question` , `#progress` , `#result-message`) and `getByRole('button', { name, exact: true })` for answers, since they are created by JavaScript. Exact matching stops "Java" also matching "JavaScript".
- **Timing:** the app advances 1 second after an answer. Use Playwright's retrying `expect` assertions rather than fixed waits. To speed up full-quiz runs, optionally control time with `page.clock` (Playwright 1.45+).
- **Data-driven tests:** keep one `QUESTIONS` array (question, correct answer, a wrong answer) and loop over it for full-quiz and result-message tests.

- **Helpers:** a `playQuiz(page, numCorrect)` helper answers all five questions with a chosen number correct.
- **Config:** set `baseURL` to `http://127.0.0.1:5500`, or add a `webServer` entry so Playwright starts Live Server itself.
- **Browsers:** Chromium by default; add Firefox and WebKit projects once the suite is stable.

Test environment and test data

Environment: the app runs at `http://127.0.0.1:5500/100Projects/Quiz_Game/index.html` via Live Server. Desktop viewport (default 1280×720) plus a mobile viewport of 375×667.

Question data is fixed in `script.js`, so expected results are deterministic:

#	Question	Correct answer	Used as wrong answer
1	What is the capital of France?	Paris	London
2	Which planet is known as the Red Planet?	Mars	Venus
3	What is the largest ocean on Earth?	Pacific Ocean	Atlantic Ocean
4	Which of these is NOT a programming language?	Banana	Java
5	What is the chemical symbol for gold?	Au	Go

Result messages depend on the percentage score:

Correct answers	Percentage	Message
5	100%	Perfect! You are a genius!
4	80%	Great job! You know your stuff!
3	60%	Good effort! Keep learning!
2	40%	Not bad! Try again to improve!
0–1	0–20%	Keep studying! You'll get better!

Test cases

There are 28 cases across seven areas. Priority: **H** = must pass for release, **M** = should pass, **L** = nice to have.

ID	Area	Test	Expected result	Pri
ST-01	Start	Load the app	Only the start screen is visible, heading "QuizTime!"	H
ST-02	Start	Click Start Quiz	Start screen hides, quiz screen shows question 1	H
ST-03	Start	Check initial quiz state	Counter "1" of "5", score 0, progress bar width 0%	H
ST-04	Start	Check answer options	Four buttons: London, Berlin, Paris, Madrid, in that order	H
AN-01	Answer	Click the correct answer (Paris)	Button gets class `correct`, score becomes 1	H
AN-02	Answer	Click a wrong answer (London)	London gets class `incorrect`, Paris gets `correct`, score stays 0	H
AN-03	Answer	Click a second option after answering	Second click is ignored, score and styling unchanged	H
AN-04	Answer	Double-click the correct answer	Score increases by 1 only	M
AN-05	Answer	Check for stray feedback	Only one `incorrect` button and one `correct` button at most per question	L
NV-01	Navigation	Answer question 1 and wait	After about 1 second, question 2 appears with its own four answers	H
NV-02	Navigation	Check counter after advancing	Counter shows 2, then 3, 4, 5 on each step	H
NV-03	Navigation	Check progress through the quiz	Progress width is 0%, 20%, 40%, 60%, 80% on questions 1 to 5	M
NV-04	Navigation	Check each question's text and answers	Match the data table for all five questions	H

ID	Area	Test	Expected result	Pri
NV-05	Navigation	Answer correctly, then wrongly, alternately	Score tracks only correct answers across all questions	M
RS-01	Results	Finish the quiz	Quiz screen hides, result screen shows after the last answer	H
RS-02	Results	Answer all 5 correctly	Final score 5, max 5, "Perfect! You are a genius!"	H
RS-03	Results	Answer 4 correctly	Final score 4, "Great job! You know your stuff!"	H
RS-04	Results	Answer 3 correctly	Final score 3, "Good effort! Keep learning!"	H
RS-05	Results	Answer 2 correctly	Final score 2, "Not bad! Try again to improve!"	H
RS-06	Results	Answer 1 correctly	Final score 1, "Keep studying! You'll get better!"	M
RS-07	Results	Answer none correctly	Final score 0, "Keep studying! You'll get better!"	M
RS-08	Results	Read the result sentence	Reads "You scored 3 out of 5" with a space before "out" (expected to fail, see Known issues)	M
RT-01	Restart	Click Restart Quiz from results	Goes straight to question 1 of a new run	H
RT-02	Restart	Check state after restart	Score 0, counter 1, progress 0%, no `correct` / `incorrect` classes	H
RT-03	Restart	Complete a second full run	Second run scores correctly (5/5 with all correct answers)	M
RP-01	Responsive	Run the quiz at 375×667	Answers visible and tappable, no horizontal scroll	M
RP-02	Responsive	Check card at 1280×720	Card centered and fully visible	L
KB-01	Keyboard	Tab to an answer and press Enter	Answer is selected, same result as a click	L

Known issues and risks

Reading `index.html` and `script.js` turned up two defects and one risk that the tests should confirm or guard against.

Ref	Finding	Impact	Related test
K-01	The result sentence has no space between `</span>` and "out of", so it renders "You scored 3out of 5"	Visible text bug	RS-08
K-02	Progress is set from the question index, so it reaches only 80% on question 5 and never shows 100%	Bar looks incomplete at the end	NV-03
K-03	Screen visibility relies on the `.active` class in CSS, so hidden-state assertions depend on `display: none`	Tests may need `not.toHaveClass(/active/)` instead	ST-01

Entry and exit criteria

Entry: the app loads at the Live Server URL, Playwright and its browsers are installed, and `index.html`, `script.js` and `style.css` are unchanged since the last run.

Exit (release-ready):

- All High-priority cases pass
- At least 90% of Medium cases pass, with any failure logged
- RS-08 and NV-03 either pass or are accepted as known issues K-01 and K-02
- No flaky tests across three consecutive runs

Later additions: screenshot comparison with `toHaveScreenshot()`, Firefox and WebKit runs, and an accessibility pass using `@axe-core/playwright`.